

CUSTOMER MILESTONE BRIEF

MS #25 - Reproducible PL/I Build Attestation

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.25.0, 0.25.1
Date	2026-08-27 to 2026-08-27
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #25 - Reproducible PL/I Build Attestation converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Makes the bounded PL/I candidate build reproducible and binds the artifact to its exact source and toolchain evidence.

Why this matters to a customer

Makes the bounded PL/I candidate build reproducible and binds the artifact to its exact source and toolchain evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a JDK-17-only reproducible builder for the bounded PL/I modernization service JAR.
- Added five deterministic execution checks with a JUnit-compatible XML report.
- Added a content-addressed dependency inventory and CycloneDX 1.5 SBOM.
- Added SLSA-shaped provenance binding the clean source commit, compiled JAR, test report, SBOM, dependency inventory, and MS #22 behavior evidence.
- Added asymmetric RSA development signatures with a hard non-release boundary and pure-Python verification against a pinned public trust anchor.
- Added GitHub workload-identity build and SBOM attestations using repository OIDC.

- Made missing, stale, tampered, foreign-workflow, replayed, or incorrectly signed build evidence demote PL/I development readiness.
- Added tamper tests for the JAR, JUnit XML, dependency inventory, commit, workflow, and release overclaim while keeping live PL/I equivalence and production readiness blocked.
- Made deterministic PL/I artifact verification survive a squash merge without weakening the committed source-tree, signature, workflow, or artifact bindings.
- Preserved exact receipt and attestation comparison whenever the recorded pre-evidence commit is reachable; otherwise compare the four portable build products byte-for-byte.

How it differs from earlier work

MS #25 builds on MS #24 (PL/I Discovery Conformance Lab). The earlier milestone established its own bounded capability; this milestone adds reproducible pl/i build attestation while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use reproducible pl/i build attestation as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.25.0, 0.25.1

- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.